# M1 Model Specification and Research Question

## Differentiable Self Calibration for Fourier Ptychography

Team number: Team 25 | Team members: Abhishek Jain, Javadizadeh Saeed, Raymound Yao, Aarya K Anup

**Scope.** This first-draft specification isolates one calibration question: whether a differentiable Fourier-ptychographic reconstruction can recover a global LED-array translation and rotation well enough to improve the reconstructed complex object. The deliberately small synthetic case makes the comparison falsifiable and executable before adding aberrations or experimental data.

### 1 Research question

**Question.** For a synthetic Fourier-ptychographic microscope reconstructing one fixed 256 x 256 complex object from a 7 x 7 LED subset at 530 nm, how much does jointly estimating global LED-array translation (dx, dy) and in-plane rotation (theta) reduce aligned object-field NRMSE relative to reconstruction with the nominal LED positions held fixed?

**A result that counts as no.** The answer is no if the joint method fails to reduce median NRMSE by at least 5% relative to the fixed-geometry baseline across 10 matched misalignment/noise seeds, or if its median NRMSE is higher.

**Decision informed.** The result determines whether geometric self-calibration belongs in the M3 reconstruction model before the team spends effort on pupil-aberration estimation and experimental validation.

**Overlap.** Other teams may study computational imaging or calibration; this question is separated by its global rigid LED-pose intervention, matched fixed-versus-joint comparison, synthetic 256 x 256 data slice, and aligned complex-object NRMSE QoI.

### 2 Conceptual model and scales

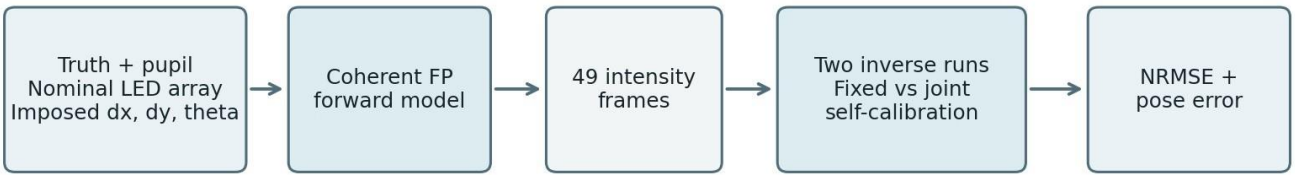



*Figure 1. Matched synthetic experiment and comparison pathway.*

**Spatial scale.** The model resolves a 256 x 256 object grid at 1 micrometer per pixel (256 micrometer square field), a pupil sampled on the same Fourier grid, and 49 illumination angles. The LED board is lumped to a rigid 2D array at a fixed 90 mm object distance.

**Temporal scale.** One static acquisition stack is modeled; optical propagation is steady-state and time is not resolved. Only optimizer iteration (target 300 steps) changes the reconstruction state.

## 3 State parameter input and output inventory

*An asterisk marks quantities treated as uncertain in the first UQ study. Values are provisional synthetic choices and will be checked against the selected reference implementation at M2.*

| Role | Symbol | Meaning | Units | Initial value or range | Source |
|---|---|---|---|---|---|
| **State** | O_hat | Estimated complex object transmission on 256 x 256 grid | relative amplitude and rad | uniform amplitude; zero phase | initialized by model |
| **State** | dx_hat | Estimated global LED-array x translation | mm | 0; constrained [-1.5, 1.5] | optimizer state |
| **State** | dy_hat | Estimated global LED-array y translation | mm | 0; constrained [-1.5, 1.5] | optimizer state |
| **State** | theta_hat | Estimated in-plane LED-array rotation | degree | 0; constrained [-3, 3] | optimizer state |
| **Input*** | I_m | Measured or synthetic intensity frame for LED m | normalized intensity | 49 frames; noise 0-2% of peak | forward model; uncertain noise |
| **Input*** | dx, dy | True translation used only to generate synthetic trials | mm | each in [-1, 1] | prescribed test distribution |
| **Input*** | theta | True rotation used only to generate synthetic trials | degree | [-2, 2] | prescribed test distribution |
| **Input*** | seed | Matched object, pose, and noise realization | dimensionless integer | 10 distinct seeds | Monte Carlo design |
| **Parameter** | lambda | Illumination wavelength | nm | 530 | fixed synthetic instance |
| **Parameter** | NA | Objective numerical aperture | dimensionless | 0.10 | fixed synthetic instance |
| **Parameter** | p_LED | Nominal LED pitch | mm | 4.0 | provisional geometry |
| **Parameter** | h_LED | LED-array distance from object | mm | 90 | provisional geometry |
| **Parameter** | P(k) | Known circular pupil for core comparison | dimensionless complex | unit magnitude inside NA | model choice; fixed |
| **Parameter** | N_iter | Maximum optimizer iterations | iterations | 300 | compute budget |
| **Parameter** | alpha | Object and pose learning rates | per iteration | 1e-2 and 1e-3 | tuning choice |
| **Output/QoI** | NRMSE | Aligned complex-object reconstruction error | dimensionless ratio | reported per seed and median | computed from known truth |
| **Output** | e_xy | Translation estimation error | mm | Euclidean pose error | computed from known truth |
| **Output** | e_theta | Rotation estimation error | degree | absolute angular error | computed from known truth |
| **Output** | R_I | Normalized intensity-data residual | dimensionless ratio | final value | computed from all frames |

## 4 Governing relations and modeling choices

| Relation | Bookkeeping or constitutive choice |
|---|---|
| r_m = R(theta) r_m,nom + [dx, dy]; k_m = (2 pi / lambda) [x_m, y_m] / sqrt(x_m^2 + y_m^2 + h_LED^2) | Bookkeeping: rigid-coordinate transform and illumination wave-vector geometry. Choice: one global rigid pose replaces independent per-LED errors; this is the intervention being tested. |
| psi_m = F^-1 { P(k) O(k - k_m) }; I_hat_m = \|psi_m\|^2 | Bookkeeping: once coherent scalar Fourier optics is adopted, illumination shifts the object spectrum, the pupil selects a band, and camera intensity is magnitude squared. Choice: thin 2D object, plane-wave LED illumination, and known pupil. |
| L = mean_m \|\|I_hat_m - I_m\|\|_2^2 / \|\|I_m\|\|_2^2 + gamma R(O_hat) | Bookkeeping: the residual compares each predicted frame with its matched input. Choice: squared normalized residual and a weak smoothness penalty R; these imply approximately Gaussian residuals and stabilize an underdetermined inverse problem. |
| z_(t+1) = Adam(z_t, grad L), where z = (O_hat, dx_hat, dy_hat, theta_hat) | Constitutive/algorithm choice: Adam and its learning rates are selected update rules, not physical laws. The same object initialization and stopping rule are used for both comparison arms. |
| NRMSE = min_c \|\|c O_hat - O_true\|\|_2 / \|\|O_true\|\|_2, for complex scalar c | Measurement choice: minimizing over c removes the unavoidable global scale/phase ambiguity so the QoI measures reconstruction structure rather than an arbitrary gauge. |

## 5 Assumptions and neglected phenomena

- Thin, static 2D transmissive object. If it fails, add axial slices as new states and move to a multislice propagation paradigm.
- Quasi-monochromatic, spatially coherent illumination. If it fails, add spectral width/coherence parameters or use an incoherent mixture model.
- One rigid LED translation and rotation explains misalignment. If it fails, add per-LED position states or a higher-order board deformation model.
- The pupil is known in the core comparison. If it fails, add low-order Zernike coefficients as states and test identifiability against pose.
- Normalized residuals are approximately independent with constant variance. If it fails, replace the squared loss with a Poisson or heteroscedastic likelihood (different constitutive choice).

**Deliberately neglected.** Chromatic aberration is omitted; failure requires wavelength-dependent pupil states. Per-LED brightness variation is omitted; failure requires one gain parameter per LED or measured flat-field inputs.

## 6 Paradigm and planned evidence

**Paradigm.** A hybrid deterministic inverse model combines a steady-state Fourier-optics forward operator with a discrete gradient-based optimizer; a Monte Carlo outer loop supplies 10 matched uncertainty trials. This directly represents the measured intensity formation while keeping calibration variables differentiable. A full Maxwell PDE model is rejected for M1-M3 because the question concerns system-level illumination geometry, and resolving every lens and free-space wavelength would add cost without improving the selected QoI.

**Planned QoIs.** Primary: per-seed and median aligned NRMSE, plus percent change from fixed geometry. Secondary: translation error e_xy, rotation error e_theta, final intensity residual, and iteration count to convergence.

**Verification.** Check autodifferentiated pose gradients against centered finite differences on a 64 x 64 reduced case (target relative discrepancy below 1e-4), then test the zero-misalignment limiting case and repeat at 128 x 128 and 256 x 256 to check QoI convergence.

**Validation.** Synthetic validation compares recovered pose and object directly with known truth. Experimental validation is not yet direct because true complex object transmission is unavailable; it would require an independently measured LED pose and a calibrated amplitude/phase target. Until then, reproduce one published calibration case and compare the direction and scale of improvement.

**Graduate-extension intent.** (1) Monte Carlo uncertainty quantification across pose and noise distributions with confidence intervals; (2) add low-order pupil aberration coefficients and test pose-aberration identifiability. Final extension scope will be declared at M2.

**Planned reference assets:** Tan et al. Uncertainty-aware Fourier ptychography (2025); Eckert et al. Efficient illumination angle self-calibration in Fourier ptychography (2018).